PLC & Industrial Automation — Complete Training Course Syllabus

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Course overview

This master syllabus covers industrial automation in 36 modules across 7 parts, from basic electrical control to IIoT, robotics and functional safety. It merges the original 22-module course with 14 new modules and expanded topics for current industry practice.

Theory • Practical • Troubleshooting • Industrial Projects

- **PLC platforms:** Siemens LOGO!, S7-200/S7-300 (legacy), S7-1200, S7-1500, Allen-Bradley (Rockwell), Mitsubishi, Delta, LS ELECTRIC, Schneider Electric and Omron (elective)
- **Programming languages:** full IEC 61131-3 — Ladder (LD), Function Block Diagram (FBD), Structured Text (ST/SCL) and Sequential Function Chart (SFC)
- **Core systems:** HMI, SCADA, VFD, servo and stepper drives, soft starters, industrial networking, IO-Link, IIoT and cybersecurity
- **Field and special systems:** sensors, encoders, pneumatics, hydraulics, process instrumentation, weighing, machine vision, RFID and industrial robots
- **Practical skills:** panel wiring, electrical CAD, calibration, diagnostics, commissioning (FAT/SAT) and industrial projects
- **Target learners:** diploma and B.Sc. engineering students, electrical/maintenance technicians and working engineers moving into automation
- **Core vs elective:** modules marked *(Elective)* can be offered as add-on courses depending on lab hardware and learner demand

Modules marked ★ **New** were added in this version. Exact hardware models, software versions and practical hours should be selected according to available training equipment.

Course structure at a glance

The course runs in 7 parts; each part builds on the one before it.

Part	Modules	Focus area
A — Foundations	01–04	Automation fundamentals, electrical control, electrical CAD and panel standards ★, PLC hardware
B — PLC Programming	05–15	IEC 61131-3 languages ★, Siemens LOGO!/S7-200/S7-300/S7-1200/S7-1500 ★, Allen-Bradley ★, Mitsubishi, Delta, LS, Schneider/Omron ★, advanced programming
C — HMI, Networking, SCADA & IIoT	16–21	HMI, industrial networking, IO-Link and remote I/O ★, SCADA, IIoT ★, cybersecurity ★
D — Drives & Motion	22–24	VFD/VFI, servo and stepper motion control, soft starter and DC converter
E — Field Devices & Special Systems	25–29	Sensors, encoders, pneumatics, hydraulics ★, instrumentation, weighing and vision ★, robotics ★
F — Applications, Power & Safety	30–33	Motor applications, power distribution and energy ★, functional safety ★, maintenance and troubleshooting
G — Engineering Practice & Projects	34–36	Project engineering and career ★, simulation and virtual commissioning, industrial projects and assessment

Full module list

No.	Module	Type
01	Introduction to Industrial Automation & Industry 4.0	Core
02	Electrical Control Systems & Panel Wiring	Core
03	Electrical Design, CAD & Panel Standards ★ New	Core
04	PLC Hardware & Operating Principles	Core
05	IEC 61131-3 Programming Languages ★ New	Core
06	PLC Programming: Siemens LOGO!	Core

No.	Module	Type
07	PLC Programming: Siemens S7-200 & S7-300 (Legacy)	Core
08	PLC Programming: Siemens S7-1200	Core
09	PLC Programming: Siemens S7-1500 & Advanced TIA Portal ★ New	Core
10	PLC Programming: Allen-Bradley / Rockwell ★ New	Core
11	PLC Programming: Mitsubishi	Core
12	PLC Programming: Delta	Core
13	PLC Programming: LS ELECTRIC	Core
14	PLC Programming: Schneider Electric & Omron ★ New	Elective
15	Advanced PLC Programming & Application Development	Core
16	HMI: Human-Machine Interface	Core
17	Industrial Networking & Communication Protocols	Core
18	IO-Link, Remote I/O & Smart Field Devices ★ New	Core
19	SCADA & Industrial Monitoring	Core
20	IIoT, Edge Computing & Cloud Integration ★ New	Core
21	Industrial Cybersecurity ★ New	Core
22	Drive Control: VFD / VFI	Core
23	Servo, Stepper & Motion Control	Core
24	Drive Control: Soft Starter & DC Converter	Core
25	Sensors, Actuators, Encoder & Electro-Pneumatics	Core
26	Hydraulics & Electro-Hydraulics ★ New	Elective
27	Process Instrumentation, Temperature Control & Calibration	Core
28	Weighing, Machine Vision, Barcode & RFID ★ New	Elective
29	Industrial Robotics & PLC–Robot Integration ★ New	Elective

No.	Module	Type
30	Industrial Motor Control & Applications	Core
31	Power Distribution, ATS & Energy Management ★ New	Core
32	Functional Safety & Machine Safety ★ New	Core
33	Industrial Safety, Maintenance & Troubleshooting	Core
34	Project Engineering, Standards & Career Development ★ New	Core
35	PLC Simulation, Digital Twin & Virtual Commissioning	Core
36	Industrial Automation Projects & Assessment	Core

Assessment and duration

Hands-on work carries the largest share of marks (30%). Practical hours should be at least 60% of contact time.

Suggested assessment weighting

Assessment category	Suggested weight
Theory and technical concepts	20%
PLC programming and simulation	25%
Hands-on wiring and hardware practice	30%
Industrial project, documentation and presentation	20%
Continuous assessment (lab reports, attendance, quizzes)	5%

Suggested duration (approximate)

Part	Modules	Theory (h)	Practical (h)	Total (h)
A — Foundations	01–04	16	24	40
B — PLC Programming	05–15	40	80	120

Part	Modules	Theory (h)	Practical (h)	Total (h)
C — HMI, Networking, SCADA & IIoT	16–21	22	38	60
D — Drives & Motion	22–24	12	24	36
E — Field Devices & Special Systems	25–29	20	34	54
F — Applications, Power & Safety	30–33	14	26	40
G — Engineering Practice & Projects	34–36	10	40	50
Total	**36**	**134**	**266**	**400**

Suggested delivery formats:

- **Full professional course:** all 36 modules, about 400 hours (6 months at 16 h/week)
- **Core course:** core modules only, about 320 hours
- **Short courses:** each part (or a single brand module) can run on its own as a 1–4 week certificate course

This weighting and hour plan is a recommendation. Adjust it to course duration, learner experience and available training hardware.

Part A — Foundations (Modules 01–04)

Module 01 — Introduction to Industrial Automation & Industry 4.0

- Discussion about the training syllabus, course objectives and learning outcomes
- Introduction to automation and control systems
- Open-loop and closed-loop control systems
- Classification of control systems: manual, automatic, sequential and process control
- Discrete, continuous and batch processes
- Introduction to PLC, PAC, DCS, SCADA, HMI, VFD and industrial controllers
- Applications of industrial automation in manufacturing and process industries (textile, garments, pharma, food & beverage, cement, power, water treatment)
- Basic electrical and electronic concepts for automation
- Introduction to sensors, actuators and industrial control devices
- Industrial automation architecture and signal flow
- Automation pyramid (ISA-95 levels 0–4): field, control, supervisory, MES and ERP
- Industry 4.0 overview: IIoT, digital twin, smart factory and data-driven maintenance

- Overview of industrial communication systems and automation standards (IEC, ISA, NFPA)
- Reading P&ID (Piping & Instrumentation Diagram) symbols and instrument tags
- Introduction to industrial safety and technical documentation

Module 02 — Electrical Control Systems & Panel Wiring

- Electrical symbols (IEC and ANSI), control diagrams and wiring diagrams
- NO/NC contacts, push buttons, selector switches, pilot lamps and emergency stop
- Relay, contactor, overload relay, MCB, MCCB, RCCB/ELCB and control fuse
- Timer relays, phase-failure relays and motor protection circuit breakers (MPCB)
- Start/Stop and self-holding circuits
- Forward/Reverse motor control and electrical interlocking
- DOL and Star-Delta starter circuits
- Control transformer, SMPS and 24 V DC power supply
- Control panel layout, DIN rail, cable ducts, terminal blocks, wire numbering, ferrules and cable identification
- Wire sizes, colour codes, lugs, crimping tools and cable glands
- Single-line diagrams and electrical schematic diagrams
- Earthing, grounding, short-circuit and overload protection
- PLC input/output field wiring
- Practical panel wiring, testing (continuity, insulation resistance with a megger) and fault tracing

Module 03 — Electrical Design, CAD & Panel Standards ★ New

- Introduction to electrical CAD for automation
- EPLAN Electric P8 (or AutoCAD Electrical) project setup, page types and symbol libraries
- Drawing power circuits, control circuits and PLC I/O pages
- Automatic wire numbering, cross-referencing and terminal diagrams
- Cable lists, terminal plans and bill of materials (BOM) generation
- Panel layout drawing: component placement, ducting and DIN-rail arrangement
- Panel design standards: IEC 60204-1 (machine electrical equipment) and IEC 61439 (LV switchgear assemblies)
- Cable sizing, voltage-drop calculation and breaker/fuse selection
- Panel heat-load calculation and cooling (fans, filters, panel AC)

- IP ratings, panel enclosure selection and segregation forms
- Component selection from manufacturer catalogues
- Project costing: BOM, labour estimation and quotation preparation
- Practical: complete drawing set for a motor control panel with PLC

Module 04 — PLC Hardware & Operating Principles

- Introduction to PLC, applications, advantages and limitations
- PLC types: compact, modular, rack-mounted, safety PLC and soft PLC/PAC
- Block diagram and working principle of a PLC
- PLC scan cycle: input scan, program execution and output update
- CPU, memory, processor and internal registers
- Digital input/output modules and internal circuit diagrams
- Relay, transistor and triac output types
- Analog input/output modules
- 0–10 V and 4–20 mA signals
- Special modules: high-speed counter, pulse output, temperature (RTD/thermocouple) and communication modules
- Distributed and remote I/O concepts
- Communication interfaces and PLC power supply
- Electronic components used in PLC modules (optocoupler, transistor, relay, fuse)
- PNP/NPN sensors, sourcing/sinking and common-terminal wiring
- PLC addressing, memory areas, data types and retentive memory
- Number systems: binary, BCD, hexadecimal, signed integers and floating point
- Hardware selection, I/O count calculation, module expansion and diagnostics

Part B — PLC Programming (Modules 05–15)

Module 05 — IEC 61131-3 Programming Languages ★ New

- IEC 61131-3 standard: software model, POUs (programs, functions, function blocks) and tasks
- Ladder Diagram (LD): rungs, contacts, coils and power flow
- Function Block Diagram (FBD): logic blocks, signal flow and block chaining
- Structured Text (ST/SCL): variables, IF/CASE, FOR/WHILE loops, arrays and expressions

- Sequential Function Chart (SFC) / GRAFCET: steps, transitions, actions, parallel and alternative branches
- Instruction List (IL): overview only (deprecated in the current edition)
- Elementary and derived data types, structures (UDT) and arrays
- Choosing the right language for the application
- Converting relay logic to ladder logic and ladder to ST
- Hands-on with CODESYS (free, vendor-neutral IEC 61131-3 environment)
- Practical: the same traffic-light sequence written in LD, FBD, ST and SFC

Module 06 — PLC Programming: Siemens LOGO!

- Introduction to Siemens LOGO! families, hardware, cables and software
- LOGO! Soft Comfort installation and project configuration
- PLC connection, program download/upload and online monitoring
- Ladder Diagram (LAD) and Function Block Diagram (FBD)
- Bit logic operations: contacts, coils, AND, OR and NOT
- Set/Reset, RS latch, pulse relay and edge detection
- Timers: on-delay, off-delay, pulse and retentive timing functions
- Weekly and yearly timer (time-switch) functions
- Counters: up, down and up/down counting
- Asynchronous timing and shift-register functions
- Digital input/output programming
- Analog input/output configuration
- Analog comparator and analog threshold functions
- Text display and message configuration
- LOGO! web server and Ethernet communication (LOGO! 8)
- Practical applications: automatic lighting, pump control, staircase light and traffic-light sequence
- Simulation, testing and troubleshooting

Module 07 — PLC Programming: Siemens S7-200 & S7-300 (Legacy)

S7-200

- Introduction to S7-200 CPU families, expansion modules, cables and software
- STEP 7-Micro/WIN installation and project configuration
- PLC communication, program upload/download and online monitoring
- Ladder Diagram programming

- Bit logic operations: contacts, coils, Set/Reset and logic instructions
- Timers, counters, edge detection and pulse operations
- Digital input/output configuration and wiring
- Analog input/output configuration and scaling
- Memory areas: I, Q, M, V, SM, AI and AQ
- Word processing, data transfer and MOVE instructions
- Compare, arithmetic and mathematical operations
- Integer, double-integer and real-number processing
- High-Speed Counter (HSC) configuration and applications
- TD-200 wizard and text-display configuration
- Interrupt routines and high-speed functions
- Creating and using MWL/AWL files where applicable to the software workflow
- Practical projects: motor control, counting and analog monitoring
- Program backup, restoration and fault diagnosis

S7-300 (maintenance focus)

- S7-300 rack, CPU, signal modules and MPI/PROFIBUS interfaces
- SIMATIC Manager (STEP 7 Classic) project structure and hardware configuration (HW Config)
- Online upload, monitoring and diagnostic buffer reading
- Migrating S7-300 projects to TIA Portal / S7-1500

Module 08 — PLC Programming: Siemens S7-1200

- Introduction to S7-1200 CPU families, models and expansion modules (SM, SB, CM)
- Hardware configuration and CPU/module selection
- TIA Portal installation, project creation and device configuration
- Ethernet connection, programming interface and IP configuration
- Ladder Diagram (LAD), Function Block Diagram (FBD) and Structured Control Language (SCL)
- PLC addressing: I, Q, M, DB and symbolic tags
- Digital input/output wiring, configuration and testing
- PLC scan cycle, memory and data types
- Bit logic, contacts, coils, Set/Reset and edge detection
- TON, TOF, TP and TONR timers; CTU, CTD and CTUD counters
- MOVE, compare, arithmetic and data conversion instructions
- Organization Blocks (OB), Functions (FC), Function Blocks (FB) and Data Blocks (DB)

- Cyclic interrupt, hardware interrupt and startup OBs
- Analog input/output configuration, scaling (NORM_X, SCALE_X) and threshold control
- High-Speed Counter (HSC) and encoder interfacing
- PTO/PWM pulse output and Motion Control technology objects (axis, homing, MC_MoveAbsolute)
- PID Compact configuration, commissioning and auto-tuning
- PROFINET communication and PLC–HMI integration
- PLC web server and data logging to memory card
- Online monitoring, watch/force tables, diagnostics, backup and restoration
- PLCSIM simulation
- Practical projects: motor control, sensor counting, analog monitoring and VFD control

Module 09 — PLC Programming: Siemens S7-1500 & Advanced TIA Portal ★ New

- S7-1500 CPU families, ET 200SP/ET 200MP distributed I/O and module selection
- Advanced hardware configuration, PROFINET topology and device names
- Optimized block access, UDTs, multi-instance FBs and libraries
- Advanced SCL programming: arrays, structures, loops and indirect addressing
- GRAPH (SFC) programming for sequential machines
- System diagnostics, ProDiag and alarm generation
- Know-how protection, access levels and copy protection
- Trace function and performance analysis
- Integrated motion control (positioning and synchronous axes overview)
- Siemens WinCC Unified / Comfort panel integration
- PLCSIM Advanced for virtual commissioning
- Practical: modular machine program built from reusable library blocks

Module 10 — PLC Programming: Allen-Bradley / Rockwell ★ New

- Rockwell PLC families: Micro800, MicroLogix (legacy), CompactLogix and ControlLogix
- Software: Connected Components Workbench (CCW) for Micro800 and Studio 5000 Logix Designer for Logix controllers
- RSLinx Classic / FactoryTalk Linx communication drivers and IP setup (BOOTP/DHCP)
- Tag-based addressing, controller vs program tags and data types
- Tasks, programs and routines (continuous, periodic, event)
- Ladder logic: XIC, XIO, OTE, OTL/OTU, ONS
- Timers (TON, TOF, RTO) and counters (CTU, CTD, RES)

- MOV, COP, compare, compute (CPT) and math instructions
- User-Defined Types (UDT) and Add-On Instructions (AOI)
- Analog I/O configuration and scaling (SCP)
- Produced/consumed tags and EtherNet/IP I/O
- PowerFlex drive integration over EtherNet/IP
- PanelView / FactoryTalk View ME overview
- Online edits, forcing, fault handling and controller fault codes
- Practical projects: motor control, conveyor sequencing and analog tank control

Module 11 — PLC Programming: Mitsubishi

- Introduction to Mitsubishi PLC families (FX3U, FX5U iQ-F, Q and iQ-R series), CPUs and expansion modules
- Programming cables, communication interfaces and compatible software
- GX Developer, GX Works2 and GX Works3 overview according to PLC compatibility
- Project creation, PLC connection, program transfer and monitoring
- Ladder programming and device addressing (X, Y, M, D, T, C, S)
- Contacts, coils, logic operations and Set/Reset
- Timers, counters, internal relays, latching and interlocking
- STL (step ladder) programming for sequential control
- Digital input/output configuration and wiring
- Data registers, word operations and arithmetic instructions
- Compare, MOVE and basic analog processing
- Analog input/output configuration and scaling
- High-speed counter and positioning instructions overview
- Program monitoring, diagnostics, backup and restoration
- GX Simulator / simulation functions
- Practical projects: motor control, sequential control and automatic counting

Module 12 — PLC Programming: Delta

- Introduction to Delta DVP, AS and AH PLC families
- CPU, expansion modules, cables and communication interfaces
- WPLSoft and ISPSoft installation and configuration according to PLC compatibility
- Project creation, download, upload and online monitoring
- Ladder programming and device addressing
- Contacts, coils, logic operations, Set/Reset and latching

- Timers, counters, interlocking and edge detection
- Step ladder (STL) sequential programming
- Digital input/output configuration and wiring
- Data registers, word processing and arithmetic operations
- Compare, MOVE and data conversion instructions
- Analog input/output configuration, scaling and threshold control
- High-Speed Counter (HSC) and high-speed pulse-output overview
- PLC communication and Modbus configuration (PLC–VFD, PLC–HMI)
- Delta DOP HMI integration overview
- Fault diagnosis, program backup and restoration
- Practical projects: motor control, sensor counting and analog control

Module 13 — PLC Programming: LS ELECTRIC

- Introduction to LS PLC families (XGB, XGK, XGI), CPUs and expansion modules
- Programming cables, communication interfaces and compatible software
- XG5000 and relevant programming environments
- Project creation, download, upload and online monitoring
- Ladder programming and device addressing
- Contacts, coils, logic operations, Set/Reset and latching
- Timers, counters, edge detection and interlocking
- Digital input/output configuration and wiring
- Data registers, MOVE, compare and arithmetic operations
- Analog input/output configuration and scaling
- Communication configuration (XG5000 / XG-PD) and Modbus overview
- Program diagnostics, backup and troubleshooting
- Practical projects: motor control, sequential operation and sensor automation

Module 14 — PLC Programming: Schneider Electric & Omron ★ New *(Elective)*

Schneider Electric

- Modicon M221, M241/M251 and M340 overview
- EcoStruxure Machine Expert – Basic (M221) and Machine Expert (M241/M251)
- Project creation, addressing, ladder and ST programming
- Altivar drive and Harmony HMI integration over Modbus

Omron

- CP1/CP2 and NX/NJ series overview
- CX-Programmer (CP series) and Sysmac Studio (NX/NJ)
- Device addressing, ladder programming, timers and counters
- EtherCAT basics with Omron NX/NJ controllers

Common

- Practical: port a motor-control program from Siemens to Schneider or Omron
- Comparison of brand instruction sets and addressing styles

Module 15 — Advanced PLC Programming & Application Development

- PLC scan time and execution behaviour
- Data types, memory allocation and register management
- Advanced timer and counter applications
- Rising-edge, falling-edge and one-shot logic
- Pulse generation and event handling
- Sequencing, step-by-step control and state-machine concepts
- SFC-based machine sequencing with modes: Auto, Manual, Step, Homing and Emergency
- Interlocking, permissive conditions and fault-reset logic
- Alarm handling: first-out alarms, alarm latching and acknowledgement
- Data transfer, comparison, arithmetic and conversion
- Analog scaling and engineering-unit conversion
- Indirect addressing, arrays, FIFO/LIFO and recipe data handling
- High-speed counting and encoder applications
- Basic PID control implementation
- PLCopen Motion Control function blocks overview
- Structured Text (ST) and Function Block Diagram (FBD) for complex logic
- Reusable program blocks, naming conventions and documentation
- Program version control (Git basics, change logs and code comparison)
- Program simulation, testing, commissioning and maintenance

Part C — HMI, Networking, SCADA, IIoT & Cybersecurity (Modules 16–21)

Module 16 — HMI: Human-Machine Interface

- Introduction to HMI and its role in industrial automation
- Main components: LCD touchscreen, CPU, memory and power supply
- HMI hardware families (Weintek, Siemens KTP/TP Comfort, Delta DOP), communication ports and selection
- EasyBuilder Pro (Weintek) installation and project creation
- Siemens WinCC (TIA Portal) Basic/Comfort panel configuration
- HMI programming, screen design and navigation
- HMI design best practice (ISA-101): high-performance graphics, colour use and screen hierarchy
- Buttons, indicators, numeric displays and data-entry objects
- Tag addressing and PLC register mapping
- PLC–HMI communication configuration
- Alarm, event, trend and data-logging functions
- Recipe management, user levels and password protection
- Scripts/macros for calculations and screen logic
- Remote HMI access (VNC, web server, mobile app) overview
- Motor status, frequency setting and fault display
- HMI simulation, download, backup and troubleshooting
- Practical implementation: PLC + HMI + VFD motor control

Module 17 — Industrial Networking & Communication Protocols

- Fundamentals of industrial communication and network topology (bus, star, ring, line)
- OSI model basics for industrial networks
- Serial communication: RS-232, RS-422 and RS-485
- Ethernet basics, IP address, subnet mask, gateway and VLAN concepts
- Baud rate, parity, stop bits and device addressing
- Modbus RTU and Modbus TCP (function codes, register types and addressing offsets)
- PROFIBUS DP and PROFINET (IO controller/device, GSD/GSDML files)
- EtherNet/IP and CIP basics
- EtherCAT overview for high-speed motion
- CANbus and CANopen overview

- Industrial Ethernet switches (managed/unmanaged), ring redundancy (MRP) and network cabling (M12, RJ45, fibre)
- PLC-to-PLC communication (S7 communication, PUT/GET, produced/consumed)
- PLC–HMI and PLC–VFD communication
- Protocol gateways and converters (Modbus-to-PROFINET, serial-to-Ethernet)
- Register mapping, data exchange and communication configuration
- Network diagnostic tools: ping, Wireshark, PRONETA and Modbus Poll/Slave
- Communication timeout, wiring errors, termination resistors and network fault diagnosis
- Introduction to OPC UA and industrial data integration

Module 18 — IO-Link, Remote I/O & Smart Field Devices ★ New

- Limits of conventional discrete and analog wiring
- IO-Link system: master, device, port classes (A/B) and 3-wire M12 cabling
- IODD files, device parameters and process data
- IO-Link master configuration on PROFINET/EtherNet/IP
- Smart sensors: distance, pressure, temperature and colour sensors over IO-Link
- Device replacement, parameter server and diagnostics
- Remote and distributed I/O (ET 200SP, Point I/O, Modbus I/O islands)
- Valve terminals (valve islands) on fieldbus
- Practical: IO-Link sensor reading and parameter change from the PLC/HMI

Module 19 — SCADA & Industrial Monitoring

- Introduction to SCADA architecture and industrial applications
- Difference between HMI and SCADA
- PLC, RTU, server, client and communication architecture
- SCADA software overview: Siemens WinCC, Ignition, AVEVA/Wonderware InTouch and FactoryTalk View SE
- Project configuration, communication drivers and tag configuration
- PLC data acquisition
- Process graphics, mimic screens and equipment status
- Alarm management (ISA-18.2): priorities, shelving, acknowledgement and alarm flood prevention
- Event logging and audit trail
- Real-time trends and historical data logging

- Production monitoring, OEE (Overall Equipment Effectiveness) calculation and report generation
- Scripting basics for SCADA (VB/Python/JavaScript depending on software)
- User authentication and access control
- Web and mobile SCADA clients
- Introduction to OPC UA and SQL database integration
- Redundancy concepts: redundant servers and networks
- SCADA communication troubleshooting
- Practical project: monitoring a tank, pump or conveyor system

Module 20 — IIoT, Edge Computing & Cloud Integration ★ New

- Industry 4.0 and IIoT architecture: sensor → PLC → edge → cloud
- OPC UA in depth: server/client setup, information model, security policies
- MQTT: broker, publish/subscribe, topics, QoS and Sparkplug B overview
- Node-RED: flows, PLC drivers (S7, Modbus, OPC UA), dashboards and alerts
- Edge gateways and industrial PCs (IoT2050, Raspberry Pi industrial, Teltonika)
- Time-series databases and dashboards (InfluxDB, Grafana)
- Cloud platforms overview (AWS IoT, Azure IoT, ThingsBoard)
- Remote access and remote PLC maintenance via secure VPN routers (eWON, Teltonika, SIMATIC RTU)
- GSM/SMS and email alarm notification
- Predictive maintenance basics: vibration, temperature and current monitoring
- Python basics for reading PLC data (python-snap7, pymodbus)
- Practical: PLC data published over MQTT to a live web/mobile dashboard

Module 21 — Industrial Cybersecurity ★ New

- Why OT security differs from IT security
- Real incidents and lessons learned (Stuxnet, Triton, ransomware in plants)
- IEC 62443 overview: zones, conduits and security levels
- Network segmentation, DMZ and firewalls between IT and OT
- PLC protection: passwords, access levels, know-how protection and disabling unused services/ports
- Secure remote access and user account management
- Patch management, backups and recovery planning
- Secure use of USB drives and engineering laptops

- OPC UA and MQTT security (certificates, TLS)
- Practical: hardening a PLC + HMI network and checking it with a basic security checklist

Part D — Drives & Motion (Modules 22–24)

Module 22 — Drive Control: VFD / VFI

- Introduction to Variable Frequency Drive (VFD) / Variable Frequency Inverter (VFI)
- Block diagram and working principle of a VFD
- Main components: rectifier, DC bus, inverter bridge (IGBT) and control circuit
- Three-phase induction motor fundamentals and torque–speed characteristics
- Control methods: V/f, sensorless vector and closed-loop vector control
- VFD sizing: motor kW/current, overload duty and derating
- VFD power wiring, motor terminals and control terminals
- VFD installation, earthing, shielded motor cable and EMC practice
- Input reactors, output (dV/dt) filters, EMC filters and harmonics overview
- Braking: DC braking, dynamic braking resistor and regenerative units
- Keypad operation and parameter programming
- Frequency, acceleration/deceleration time and motor nameplate settings
- Motor auto-tuning
- Start/Stop, Forward/Reverse and speed control
- Local/Remote control and external potentiometer operation
- Digital input and analog input configuration
- Preset speed, multi-speed operation and jog control
- Built-in PID for pump/fan constant-pressure control
- PLC + VFD integration using digital and analog signals
- Modbus communication between PLC and VFD (control word, status word, frequency reference)
- PROFINET/EtherNet/IP drive control overview
- Drive software tools for parameter backup (e.g. VFDSoft, SINAMICS Startdrive)
- Fault codes, overload protection and troubleshooting
- Practical exercises using Delta MS300, Siemens SINAMICS V20/G120 or a compatible training drive

Module 23 — Servo, Stepper & Motion Control

Servo systems

- Introduction to servo systems and industrial applications
- Main components: servo drive, servo motor, encoder and controller
- Servo system block diagram and feedback principle (position, speed and current loops)
- Servo motor wiring and power connections
- Servo drive parameter configuration
- Position, speed and torque control modes
- Pulse/direction control and positioning fundamentals
- Encoder feedback, homing and electronic gearing overview
- Servo control over fieldbus (EtherCAT, PROFINET, Modbus) overview
- Gain tuning: auto-tuning, inertia ratio and resonance suppression
- PLC-to-servo interface and motion-sequence programming
- Servo alarm identification and troubleshooting

Stepper systems ★ New

- Stepper motor types: permanent magnet, hybrid, 2-phase and 3-phase
- Step angle, microstepping and holding torque
- Stepper driver wiring and DIP-switch settings (current, microstep)
- Open-loop vs closed-loop (hybrid servo) steppers
- PLC pulse output (PTO) control of stepper drivers

Motion control applications

- Point-to-point positioning, linear interpolation and XY tables
- Electronic cam and flying-saw/cut-to-length overview
- Lead screw, belt and rack-and-pinion mechanical calculations
- Practical exercises: position control, repetitive motion sequences and XY pick-and-place

Module 24 — Drive Control: Soft Starter & DC Converter

- Introduction to soft starters and their applications
- Comparison: DOL vs Star-Delta vs soft starter vs VFD
- Main components and block diagram of a soft starter
- SCR/thyristor-based voltage control principle
- Soft starter wiring, motor connection (in-line and inside-delta) and parameter settings

- Ramp-up, ramp-down and current-limit settings
- Pump control (anti-water-hammer) stop settings
- Bypass contactor and motor-protection functions
- Soft starter fault diagnosis and troubleshooting
- Introduction to DC converters and their block diagrams
- AC-to-DC conversion, rectification and DC voltage control
- Basic DC motor control and industrial converter applications
- Converter protection and electrical safety

Part E — Field Devices, Instrumentation, Vision & Robotics (Modules 25–29)

Module 25 — Sensors, Actuators, Encoder & Electro-Pneumatics

Sensors and actuators

- Introduction to industrial sensors and actuators
- Inductive, capacitive and photoelectric proximity sensors (through-beam, retro-reflective, diffuse)
- Fibre-optic, colour, contrast and laser distance sensors
- Limit switches, reed switches and ultrasonic sensors
- Temperature, pressure, level and flow sensors
- Sensor selection: sensing range, response time, IP rating and output type
- PNP/NPN sensor wiring and PLC input interfacing
- Relay, transistor and analog output interfacing

Pneumatics and electro-pneumatics

- Compressed air generation, air treatment and FRL units
- Solenoid valves, pneumatic cylinders and directional control valves (3/2, 5/2, 5/3)
- Single-acting and double-acting cylinders, flow control and speed control
- Vacuum generators and suction cups
- Pneumatic circuits and actuator sequencing (A+B+A−B− notation)
- Electro-pneumatic circuits with relays and with PLC
- Pneumatic simulation (FluidSIM or equivalent)

Encoders

- Encoder fundamentals: incremental and absolute encoders (single-turn, multi-turn)
- Optical and magnetic encoders

- Encoder resolution, PPR/CPR and pulse frequency
- A, B and Z channels and quadrature signals
- Direction detection and position counting
- Encoder output types (push-pull, line driver, open collector) and electrical interfacing
- Absolute encoder interfaces: SSI, PROFINET and IO-Link overview
- Encoder connection to PLC High-Speed Counter (HSC)
- RPM, position, speed and length measurement
- Encoder feedback in servo systems

Practical

- Practical exercises: motor RPM measurement, conveyor length measurement and cylinder sequencing
- Sensor, actuator and encoder testing and troubleshooting

Module 26 — Hydraulics & Electro-Hydraulics ★ New *(Elective)*

- Hydraulic principles: Pascal's law, pressure, flow and force calculations
- Hydraulic power unit: pump, reservoir, filter, relief valve and accumulator
- Directional, pressure and flow control valves
- Hydraulic cylinders and motors
- Hydraulic symbols and circuit reading (ISO 1219)
- Electro-hydraulic control with solenoid valves and PLC
- Proportional and servo valves overview
- Pressure switches and transducers in hydraulic systems
- Hydraulic safety, oil maintenance and leak troubleshooting
- Practical: PLC-controlled hydraulic press or clamping circuit (trainer or simulation)

Module 27 — Process Instrumentation, Temperature Control & Calibration

- Introduction to process instrumentation and measurement
- Calibration concepts: switches, sensors, transmitters and measuring instruments
- Pressure, temperature, level and flow measurement
- Temperature sensors: RTD (PT100/PT1000), thermocouples (J, K, T, R/S) and cold-junction compensation
- Level measurement: float, hydrostatic, ultrasonic, radar and capacitive
- Flow measurement: electromagnetic, vortex, turbine, Coriolis and differential pressure
- Transmitter types, ranges, units and output signals
- 4–20 mA current loops and 0–10 V signals

- Two-wire, three-wire and four-wire transmitter connections
- Signal isolators, loop-powered displays and signal converters
- Calibration standards, zero/span adjustment and accuracy
- Pressure transmitter calibration using a hand pump
- Temperature calibration using a dry-block calibrator and RTD/TC simulator
- HART communication and HART calibrator operation
- Calibration documentation, as-found/as-left records and error calculation
- Analog signal testing and loop troubleshooting
- PLC analog input configuration and engineering-unit scaling
- Standalone temperature/PID controllers (e.g. Omron E5CC, Delta DTB): setup, alarms and SSR output
- SSR and SCR power controllers for heaters
- Control valves, positioners and actuators
- Process control loops, setpoint and process variable
- PID control fundamentals and tuning (manual, Ziegler–Nichols, auto-tune)
- Cascade and ratio control overview
- Practical calibration, testing and fault diagnosis

Module 28 — Weighing, Machine Vision, Barcode & RFID ★ New *(Elective)*

Weighing systems

- Load cell types (shear beam, S-type, compression) and strain-gauge principle
- Load cell wiring, junction boxes and excitation
- Weighing indicators/transmitters and PLC weighing modules
- Calibration with test weights, zero and span
- Batching, filling and check-weighing applications

Machine vision

- Vision system components: camera, lens, lighting and processor
- Smart cameras and vision sensors (Cognex, Keyence, SICK overview)
- Inspection tasks: presence, colour, shape, measurement and OCR
- Triggering, results and PLC communication

Identification

- Barcode and 2D (QR/DataMatrix) scanners and their PLC interfacing
- RFID readers, tags and frequency bands (LF, HF, UHF)
- Product tracking and traceability applications

Practical

- Colour/shape sorting with a vision sensor and pneumatic reject
- Weighing-based filling with load cell and PLC

Module 29 — Industrial Robotics & PLC–Robot Integration ★ New *(Elective)*

- Introduction to industrial robots and applications (welding, palletizing, pick-and-place, painting)
- Robot types: articulated (6-axis), SCARA, delta, Cartesian and collaborative (cobot)
- Robot anatomy: axes, end-effectors, grippers and tool centre point (TCP)
- Coordinate systems: world, base, tool and user frames
- Teach pendant operation, jogging and point teaching
- Basic robot programming: motion types (joint, linear, circular), I/O and loops
- Offline programming and simulation (ABB RobotStudio, FANUC ROBOGUIDE or RoboDK)
- PLC–robot interfacing: digital handshake signals and fieldbus (PROFINET/EtherNet/IP)
- Robot cell safety: fences, light curtains, safety zones and cobot standards (ISO 10218, ISO/TS 15066)
- Practical: PLC-controlled pick-and-place cell with a desktop robot arm or simulation

Part F — Applications, Power & Safety (Modules 30–33)

Module 30 — Industrial Motor Control & Applications

- Industrial motor types (induction, synchronous, PMSM, DC, single-phase) and nameplate interpretation
- Motor efficiency classes (IE1–IE4) and motor selection
- DOL, Forward/Reverse and Star-Delta starter circuits
- Motor overload, short-circuit, phase-failure, phase-reversal and thermistor (PTC) protection
- PLC-controlled motor starting and stopping
- Sequential motor operation and interlocking
- Auto/Manual mode selection
- Duty/standby pump control, alternating operation and run-hour balancing
- Multi-pump booster set with VFD and pressure transmitter
- Compressor and chiller sequencing
- Conveyor control and object counting

- Tank filling, draining and level-based automation
- Fault indication, alarms and safe restart logic
- PLC + HMI + VFD motor control
- Motor insulation testing and winding resistance checks
- Practical wiring, programming, testing and commissioning

Module 31 — Power Distribution, ATS & Energy Management ★ New

- LT power distribution: transformer, LT panel, MDB, SDB and bus-bar systems
- ACB, MCCB settings and protection coordination (selectivity)
- Automatic Transfer Switch (ATS): mains–generator changeover logic with PLC/ATS controller
- Generator controllers (Deep Sea, ComAp overview), auto-start and synchronization basics
- Power factor: causes, effects and correction
- APFC (Automatic Power Factor Correction) panel: capacitor banks, contactors and controller setup
- Harmonics from VFDs and UPS, and mitigation (reactors, filters)
- Digital multifunction energy meters and reading them over Modbus
- Energy monitoring dashboard on HMI/SCADA (kWh, kW, PF, demand)
- Energy-saving opportunities: VFDs on pumps/fans, compressed-air leaks and motor efficiency
- UPS and DC power systems for control panels
- Practical: ATS logic with PLC and energy-meter data on HMI

Module 32 — Functional Safety & Machine Safety ★ New

- Machine safety principles and the legal/standards framework
- Risk assessment (ISO 12100): hazard identification, risk estimation and risk reduction
- ISO 13849-1: Performance Level (PLr, PL a–e), categories (B, 1–4), MTTFd and DC
- IEC 62061 and IEC 61508: Safety Integrity Level (SIL) overview
- IEC 61511 and SIS (Safety Instrumented Systems) for process plants overview
- Emergency stop categories (0, 1, 2) per IEC 60204-1
- Safety devices: safety relays, light curtains, safety mats, door interlocks, two-hand controls and safety laser scanners
- Safety relay wiring: dual-channel, cross-monitoring and feedback (EDM) loops
- Safety PLC: Siemens F-CPU (Safety Integrated), PROFIsafe, Rockwell GuardLogix overview

- Safe Torque Off (STO) and safe motion functions in drives
- Validation and verification of safety functions (SISTEMA software overview)
- Practical: guard door + E-stop + light curtain safety circuit with a safety relay and STO drive

Module 33 — Industrial Safety, Maintenance & Troubleshooting

- Electrical hazards, arc flash and safe working practices
- Personal Protective Equipment (PPE) and insulated tools
- Lockout/Tagout (LOTO) and electrical isolation
- Emergency stop and machine-safety principles (summary of Module 32)
- Overload, short circuit, earth fault and grounding
- Electrical noise, shielding and signal interference
- PLC CPU, I/O module and power-supply fault diagnosis
- Sensor, actuator and field-wiring troubleshooting
- HMI, VFD and communication fault diagnosis
- Multimeter, clamp meter, insulation tester, loop calibrator and oscilloscope use
- Thermal imaging camera for panel inspection
- Reading wiring diagrams and tracing faults
- Systematic troubleshooting method (half-split, symptom–cause analysis) and root cause analysis (5-Why, fishbone)
- PLC program backup, restoration and version management
- Preventive, predictive and condition-based maintenance
- Spare-parts management and critical spares list
- Preventive maintenance and inspection checklists
- Commissioning documentation and maintenance reports

Part G — Engineering Practice, Simulation & Projects (Modules 34–36)

Module 34 — Project Engineering, Standards & Career Development ★ New

Project engineering

- Automation project life cycle: enquiry, URS/FDS, design, build, test, commissioning and handover
- Writing a Functional Design Specification (FDS) and I/O list
- Cause-and-effect matrices and control narratives

- ISA-88 batch control concepts: recipes, phases and units
- ISA-95 and MES basics: production orders, traceability and OEE
- FAT (Factory Acceptance Test) and SAT (Site Acceptance Test) procedures and checklists
- Loop checking, I/O checking and commissioning sequence
- As-built drawings, O&M manuals and client handover
- Project planning, time estimation and costing
- Regulated industries overview: GMP and 21 CFR Part 11 (pharma), food safety

Career development

- Automation career paths: maintenance, commissioning, design, SCADA/IIoT and system integration
- Technical CV and portfolio (projects, code samples, videos)
- Interview preparation: common PLC, VFD and troubleshooting questions
- Vendor certifications overview (Siemens, Rockwell, Mitsubishi, CODESYS)
- Freelancing and system-integrator business basics
- Technical communication and report writing

Module 35 — PLC Simulation, Digital Twin & Virtual Commissioning

- Introduction to PLC simulation and virtual training
- Simulation tools: PLCSIM, PLCSIM Advanced, GX Simulator, CODESYS SoftPLC
- Ladder logic simulation and online monitoring
- Virtual digital inputs, outputs, timers and counters
- Analog input simulation and scaling verification
- HMI screen simulation and tag testing
- 3D plant simulation with Factory I/O (or equivalent) connected to a PLC
- Digital twin concept and machine models (NX MCD, Simumatik, Emulate3D overview)
- Simulating sensor, motor and actuator behaviour
- PLC sequence testing before physical implementation
- Fault injection and troubleshooting exercises
- Simulation versus real hardware: limitations and differences
- Virtual commissioning of a small automation system

Module 36 — Industrial Automation Projects & Assessment

Core projects (all learners)

1. **PLC-based motor control** — Start/Stop, Forward/Reverse, interlocking, overload protection and fault indication
2. **PLC + HMI + VFD integration** — HMI motor control, speed setting, status monitoring and drive fault handling
3. **Automatic tank level control** — level sensing, automatic pump control, high/low-level alarms and Manual/Auto mode
4. **Conveyor and encoder automation** — sensor-based object detection, speed/position measurement, counting and sequence control
5. **Process instrumentation** — pressure transmitter calibration, 4–20 mA measurement, PLC scaling and alarm generation
6. **IIoT monitoring** — PLC data published over MQTT/OPC UA to a live web or mobile dashboard

Elective project bank (choose at least one)

7. Bottle filling and capping line (conveyor + sensors + pneumatics)
8. Colour/shape sorting system (colour sensor or vision camera + pneumatic pusher)
9. Elevator / lift model, 3–4 floors, programmed in SFC
10. Automatic car parking system with slot counting and display
11. Batch mixing plant with load cell, valves, agitator and HMI recipes
12. Boiler or water treatment plant simulation with PID loops and SCADA
13. Compressor or pump duty/standby sequencing with run-hour balancing
14. Textile dyeing machine temperature-profile control
15. Generator ATS panel with energy-meter monitoring
16. Smart greenhouse: temperature, humidity and irrigation control
17. XY pick-and-place with stepper/servo drives
18. Robot pick-and-place cell with PLC handshake and safety circuit
19. Safety-rated machine guard with light curtain, safety relay and STO

Assessment

- Practical examination and fault-finding test
- PLC program development and explanation
- Electrical wiring and commissioning assessment
- HMI/VFD communication test
- Project documentation: FDS, I/O list, EPLAN/CAD wiring diagrams and FAT checklist
- Project presentation and viva voce
- Written examination and practical evaluation

Recommended lab equipment and software

One PLC trainer per 2 learners is the target; elective modules can share a single demonstration rig.

Area	Hardware	Software	Modules
Electrical control	Contactors, relays, overloads, MCB/MPCB, push buttons, timer relays, 3-phase motor, wiring boards	—	02, 30
Electrical CAD	Engineering PCs	EPLAN Electric P8 (education) or AutoCAD Electrical	03
Siemens PLC	LOGO! 8, S7-1200 (CPU 1214C DC/DC/DC), S7-1500 with ET 200SP; S7-200/S7-300 for legacy practice	LOGO! Soft Comfort, STEP 7-Micro/WIN, SIMATIC Manager, TIA Portal + PLCSIM	06–09
Rockwell PLC	Micro820/850 or CompactLogix 5380	CCW, Studio 5000, FactoryTalk Linx	10
Other PLCs	Mitsubishi FX5U, Delta DVP/AS, LS XGB, Schneider M221, Omron CP2E	GX Works2/3, ISPSoft/WPLSoft, XG5000, Machine Expert, CX-Programmer	11–14
IEC 61131-3	Any PC	CODESYS	05, 35
HMI	Weintek cMT/MT8000, Siemens KTP700, Delta DOP	EasyBuilder Pro, WinCC (TIA), DOPSoft	16
Networking	Managed Ethernet switch, RS-485 converters, PROFINET/Modbus devices, IO-Link master + 2–3 IO-Link sensors	Wireshark, PRONETA, Modbus Poll/Slave	17, 18
SCADA & IIoT	Server PC, IoT gateway (SIMATIC IOT2050 or Raspberry Pi), 4G VPN router	WinCC / Ignition (free trial), Node-RED, Mosquitto MQTT, InfluxDB, Grafana	19–21
Drives	VFD (Delta MS300 / SINAMICS V20 or G120), servo kit, stepper motor + driver, soft starter	Drive tools (VFDSoft, Startdrive)	22–24

Area	Hardware	Software	Modules
Field devices	Inductive, capacitive, photoelectric, ultrasonic sensors; incremental and absolute encoders	—	25
Pneumatics	Compressor, FRL, 5/2 and 5/3 solenoid valves, single/double-acting cylinders	FluidSIM	25
Hydraulics	Hydraulic trainer (or simulation only)	FluidSIM Hydraulics	26
Instrumentation	Pressure transmitter, hand pump, HART calibrator, loop calibrator, RTD/TC simulator, temperature controller, SSR + heater	—	27
Weighing & vision	Load cell + weighing indicator, vision sensor, barcode/RFID reader	Vendor setup tools	28
Robotics	Desktop 6-axis or SCARA robot arm (or simulation only)	RobotStudio / ROBOGUIDE / RoboDK	29
Power & energy	Multifunction energy meter, ATS demo panel, APFC controller with capacitor bank	—	31
Safety	Safety relay, light curtain, safety door switch, E-stop, STO-capable drive	SISTEMA	32
Test instruments	Multimeters, clamp meters, insulation tester, oscilloscope, thermal camera	—	33
Simulation	Engineering PCs	Factory I/O, PLCSIM Advanced	35

Implementation notes

- PLC software, instruction sets and hardware capabilities differ by model and version. Check every practical lesson against the equipment actually available.
- Siemens S7-200 and S7-300 are legacy platforms. Teach them for maintenance only and give most hands-on time to S7-1200/S7-1500 and current networking practice.
- Teach Module 05 (IEC 61131-3) before the brand modules, so learners see each brand as a dialect of the same standard.
- Run brand modules 06–14 in parallel lab groups if trainers and hardware allow, then rotate groups.
- Encoder selection and counting reliability depend on pulse rate, output type, electrical interface and the PLC high-speed counter specifications.
- Elective modules (14, 26, 28, 29) can be delivered with simulation software where hardware is not available.
- Use free or education licences where possible: CODESYS, Node-RED, Mosquitto, Grafana, Factory I/O education and vendor trial licences.
- Keep one shared Git repository or folder of all lab programs, with version history, as course reference material.
- High-voltage wiring, motor testing and commissioning must be performed under appropriate supervision and safety procedures, with LOTO applied.
- Review the syllabus every year against new software versions and local industry demand.

What's new compared with the 22-module version

All 22 original modules are kept, renumbered into the 7-part structure, and 14 modules are new.

New module	What it adds
03 — Electrical Design, CAD & Panel Standards	EPLAN/AutoCAD Electrical, IEC 60204-1, IEC 61439, cable sizing, heat load, BOM and costing
05 — IEC 61131-3 Programming Languages	LD, FBD, ST and SFC/GRAFCET; CODESYS
09 — Siemens S7-1500 & Advanced TIA Portal	ET 200SP, GRAPH, libraries, ProDiag, PLCSIM Advanced
10 — Allen-Bradley / Rockwell	Micro800, CompactLogix, Studio 5000, AOIs, EtherNet/IP

New module	What it adds
14 — Schneider Electric & Omron	Modicon M221/M241, Machine Expert, CX-Programmer, Sysmac Studio
18 — IO-Link, Remote I/O & Smart Field Devices	IO-Link masters and devices, IODD, valve islands, distributed I/O
20 — IIoT, Edge Computing & Cloud Integration	OPC UA, MQTT, Node-RED, Grafana, VPN remote access, SMS alarms
21 — Industrial Cybersecurity	IEC 62443, segmentation, PLC hardening
26 — Hydraulics & Electro-Hydraulics	Hydraulic circuits, proportional valves, PLC control
28 — Weighing, Machine Vision, Barcode & RFID	Load cells, vision sensors, barcode and RFID tracking
29 — Industrial Robotics & PLC–Robot Integration	Robot programming, offline simulation, PLC handshake, cobot safety
31 — Power Distribution, ATS & Energy Management	ATS, generators, APFC, harmonics, energy meters
32 — Functional Safety & Machine Safety	ISO 12100, ISO 13849, SIL, safety relays, safety PLC, STO
34 — Project Engineering, Standards & Career Development	FDS, FAT/SAT, ISA-88/95, documentation, CV and interviews

Existing modules expanded:

- **01** — Industry 4.0, the ISA-95 automation pyramid and P&ID reading
- **04** — number systems, special modules and I/O sizing
- **07** — S7-300 maintenance and migration
- **08** — Motion Control technology objects, PID auto-tuning and PLCSIM
- **11–13** — STL step-ladder programming and current CPU families
- **15** — SFC machine modes, FIFO/recipes, PLCopen and Git version control
- **16** — Siemens WinCC panels and ISA-101 HMI design
- **17** — EtherNet/IP, EtherCAT, ring redundancy and network diagnostic tools
- **19** — ISA-18.2 alarm management, OEE and named SCADA platforms
- **22** — vector control, braking, EMC/harmonics and built-in PID
- **23** — stepper drives and motion applications
- **25** — electro-pneumatics and absolute encoder interfaces

- **27** — RTD/thermocouples, standalone temperature controllers and cascade control
- **33** — arc flash, thermal imaging and root-cause analysis
- **35** — Factory I/O and digital twin
- **36** — IIoT core project and a 13-project elective bank